

Paramount Diagnostic Lab

Test Details

METHODOLOGY

Liquid chromatography/tandem mass spectrometry (LC/MS-MS)

RESULT TURNAROUND TIME

4 - 10 days

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

25-Hydroxy vitamin D is used to measure vitamin D status; it includes results for total 25-hydroxy vitamin D₂ and 25-hydroxy vitamin D₃. This 25-hydroxy vitamin D assay and the alternative immunoassay test (Vitamin D, 25-Hydroxy [081950]) (/tests/081950/vitamin-d-25-hydroxy) are both CDC-certified for accuracy.¹

Custom Additional Information

The biological function of vitamin D is to maintain normal levels of calcium and phosphorus absorption. 25-Hydroxy vitamin D is the storage form of vitamin D. Vitamin D assists in maintaining bone health by facilitating calcium absorption. Vitamin D deficiency can also cause osteomalacia, which frequently affects elderly patients. Vitamin D from sunshine on the skin or from dietary intake is converted predominantly by the liver into 25-hydroxy vitamin D, which has a long half-life and is stored in the adipose tissue. The metabolically active form of vitamin D, 1,25-dihydroxy vitamin D, which has a short life, is then synthesized in the kidney as needed from circulating 25-hydroxy vitamin D. The reference interval of greater than 30 ng/mL is a target value established by the Endocrine Society.²

Specimen Requirements

Specimen

Serum (preferred) or plasma

Volume

1 mL

Minimum Volume

0.5 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube, gel-barrier tube, lavender-top (EDTA) tube, or green-top (heparin) tube

Collection Instructions

Serum or plasma must be separated from cells within 45 minutes of venipuncture. Submit serum or plasma in a plastic transport tube.

Storage Instructions

Freeze. Stable frozen for 34 months. Stable at room temperature or refrigerated for seven days.

Freeze/Thaw Cycle: Stable x6

Footnotes

1. Centers for Disease Control and Prevention (CDC). *Clinical Standardization Programs*. CDC website: <https://www.cdc.gov/clinical-standardization-programs/php/hormones/improving-performance-host.html>. (<https://www.cdc.gov/clinical-standardization-programs/php/hormones/improving-performance-host.html>.) Updated April 2024. Accessed August 2025.

2. Holick MF, Binkley NC, Bischoff-Ferrare HA, et al. Evaluation, treatment, and prevention of vitamin D deficiency: An Endocrine Society clinical practice guideline. *J Clin Endocrinol Metab*. 2011; 96(7):1911-1930. PubMed 21646368 (<http://www.ncbi.nlm.nih.gov/pubmed/21646368>)